

University Erp Report

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How to Run

First create and seed data into the two dbs

```
mysql -u root -p < sql/auth_db_create.sql
```

```
mysql -u root -p < sql/erp_db_create.sql
```

```
mysql -u root -p < sql/seed_auth.sql
```

```
mysql -u root -p < sql/seed_erp.sql
```

Write your own mysql password in src/main/java/edu/univ/erp/db/Database.java

Compile the code using

```
mvn clean package
```

Then run using

```
mvn exec:java -Dexec.mainClass="edu.univ.erp.Main"
```

Credentials

We have seeded some users in the dbs which are

Username	Password
----------	----------

admin1	admin1
--------	--------

inst2	inst2
-------	-------

inst1	inst1
-------	-------

stu1	stu1
------	------

stu2	stu2
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Report

SCREENSHOTS

Admin's Interfaces:(User name - Admin1)

University ERP System - Welcome, admin1 (ADMIN) ✓ SYSTEM ACTIVE Logout

User Management

Course Management

Section Management

System Settings

Backup/Restore

User Management

All System Users

All Registered Users (5 users)

User ID	Username	Role	Status
4	stu1	STUDENT	ACTIVE
5	stu2	STUDENT	ACTIVE
2	inst1	INSTRUCTOR	ACTIVE
3	inst2	INSTRUCTOR	ACTIVE
1	admin1	ADMIN	ACTIVE

Close

University ERP System - Welcome, admin1 (ADMIN) ✓ SYSTEM ACTIVE Logout

User Management

Course Management

Section Management

System Settings

Backup/Restore

Course ID	Code	Title	Credits	Description
63	COM101	Communication Skills	4	Course about communication skills
61	CSE101	Introduction to Programming	4	Basic programming concepts using P...
65	CSE112	Computer Organisation	4	Basics of co like assembler, simulator ..
67	CSE231	Operating System	4	Operating system in computers with c...
70	CW	Community Work	2	Mandatory community work in ngos ...
64	DES102	Introduction to HCI	4	Basics of design, ui and hci
62	ECE111	Digital Circuits	4	Basic electrical circuit analysis
66	MTH203	Maths III	4	Differential and integral calculus
69	SG	Self Growth	2	Self growth course in disciplines like c...
68	SSH201	Research Methods in Social Science a...	4	Course about quantitative and qualita...

Student's Interface (User - stu1):

University ERP System - Welcome, stu1 (STUDENT)

✔ SYSTEM ACTIVE

Logout

Course Catalog

My Enrollments

My Grades

Timetable

Transcript

Refresh

Enroll in Selected Section

Select	Section ID	Course Code	Course Title	Credits	Instructor	Day/Time	Room	Capacity	Available Seats
☐	43	COM101	Communication S...	4	inst2	Mon 12:30 PM - 2...	C203	40	39
☐	44	CSE101	Introduction to Pr...	4	inst1	Mon 11:20 AM - 1...	C302	50	49

University ERP System - Welcome, stu1 (STUDENT)

✔ SYSTEM ACTIVE

Logout

Course Catalog

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Refresh

Export CSV

Export PDF

Course	Component	Score	Max Score	Percentage	Weight
CSE101	final	39.0	50.0	78.0%	0.50%
CSE101	midterm	23.0	30.0	76.7%	0.30%
CSE101	quiz	13.0	20.0	65.0%	0.20%

University ERP System - Welcome, stu1 (STUDENT)

SYSTEM ACTIVELogout

Course Catalog

My Enrollments

My Grades

Timetable

Transcript

My Weekly Timetable

MY WEEKLY TIMETABLE
=====

Course: CSE101 - Introduction to Programming
Time: Mon 11:20 AM - 12:20 PM
Room: C302
Instructor: inst1

Course: COM101 - Communication Skills
Time: Mon 12:30 PM - 2:00 PM
Room: C203
Instructor: inst2

Instructor's Interface (User- Inst1):

University ERP System - Welcome, inst1 (INSTRUCTOR)

SYSTEM ACTIVELogout

My Sections

Grade Management

Class Statistics

Export Grades

RefreshManage GradesUpdate Enrollment Counts

Select	Section ID	Course Code	Course Title	Day/Time	Room	Capacity	Enrolled	Available	Semester	Year
☐	44	CSE101	Introduction to ...	Mon 11:20 AM -	C302	50	1	49	Monsoon	2025

My Sections

Grade Management

Class Statistics

Export Grades

Load Students

Save Grades

Compute Final Grades

Refresh

Back to Sections

Help

Enrollment ID	Student ID	Username	Roll No	Quiz (0-20)	Midterm (0-30)	Final (0-50)	Final Grade
30	4	stu1	2024100	13	23	39	75.0% (C)

Configure Grade Weights

?

Quiz Weight (%):

20

Midterm Weight (%):

30

Final Weight (%):

50

OK

Cancel

My Sections

Grade Management

Class Statistics

Export Grades

Load Statistics

CLASS STATISTICS
=====

Section: Introduction to Programming (ID: 44)

Total Students: 1

Students with Grades: 1

Capacity: 50

Currently Enrolled: 1

Available Seats: 49

QUIZ STATISTICS:

Average Score: 13.00

Maximum Score: 13.00

Minimum Score: 13.00

Standard Deviation: 0.00

Students Graded: 1

FINAL STATISTICS:

Average Score: 39.00

Maximum Score: 39.00

Minimum Score: 39.00

Standard Deviation: 0.00

Students Graded: 1

MIDTERM STATISTICS:

Average Score: 23.00

Maximum Score: 23.00

Minimum Score: 23.00

Standard Deviation: 0.00

Students Graded: 1

We used a lot of concepts of java programming which were taught in the course. While making the project we have used concepts like classes, interfaces, inheritance, swing and so on. Here is the report of our project. Making this project was enjoyable and nice but also filled with challenges like for example completing the work before the deadline. Even then it was an awesome experience and helped us learn java in even depth.

The basic structure of our project follows that given in the assignment as follows. We have made separate folders inside `src/main/java/edu/univ/erp`.

Ui folder – This is all the ui code of the project. It has the screens for admin, student, instructor and login.

Domain folder — This folder has classes for the tables in the databases like student, instructor etc so that we can make objects of whatever data is in the dbs.

Service folder — This is the functionality code. This also implements the maintenance flag case which the admin was toggling.

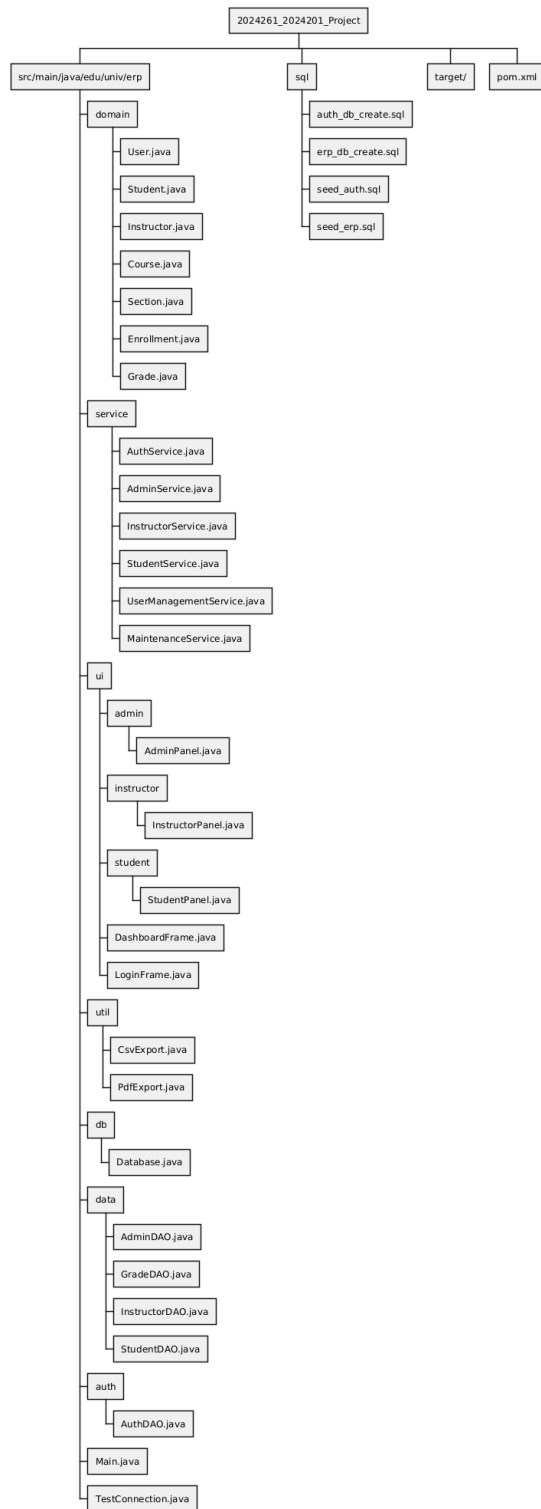
Data folder — This is the database helper for ERP DB and has the erp db code.

Auth folder — This is the database helper for Auth DB and has the auth db code.

Db folder — We have used Hikari to make connection with the databases. This just has one file which uses Hikari to connect with the dbs and use it in the database helper.

Util folder — This is the folder for CSV/PDF export. It is needed to export the grades of a student in the form of CSV or PDF.

Project directory looks like:



How role and Maintenance are enforced

We have 3 roles – student, instructor and admin. To enforce these roles, we made separate UI screens for each of them and made a column of “role” in the Auth db. When the user logs in, we identify the role of the user from the Auth db and then display that particular screen. For functionalities in each role specific to that role like for example adding a user in the admin or viewing timetable in student we have used the database helpers (in `src/main/java/edu/univ/erp/auth` or `src/main/java/edu/univ/erp/data`) to basically modify the data in the database and when needed we added the functionality in the service folder.

So maintenance is stored in the “setting” table of ERP db. The admin can toggle this to true or false. To implement this functionality, we have used simple logic. We check whether the maintenance is true or not and according to that the access changes. The admin is not affected but the student and instructor are view only. In the UI code of student and instructor we checked this access and accordingly the access changes.

Final Grade Weighting Rule

The weightage for the final grade is given by the instructor. The individual grades (like quiz, midsem etc) are stored in the “grades” table of the ERP db along with their respective weightage. Hence from this the final mark is calculated. The instructor can change the weightage of each component (quiz, midsem, final) to calculate the final percentage. The marks for quiz ranges (0-20), midsem ranges (0-30), final exam ranges from (0-50). The grade we have given is

A (90% - 100%)

B (80% - 89%)

C (70% - 79%)

D (60% - 69%)

F (0% - 59%)

based on percentage of the student.

Table list for DB

The table list was given in the assignment, we have made the tables according to that-

Auth DB

`users_auth` with columns like `user_id`, `username`, `password_hash` etc.

ERP DB

`students`(`user id`, `roll no`, `program`, `year`)

`instructors`(`user id`, `department`, ...)

courses(code, title, credits)

sections(course id, instructor id, day time, room, capacity, semester, year)

enrollments(student id, section id, status)

grades(enrollment id, component, score, final grade)

settings(key, value) with maintenance on = true/false

Video

Attached in the project's submitted zip File

Testing Summary

We have thoroughly tested the project and here is the summary of whatever we did.

First of all, the connectivity with the database is checked. If the connection is successful then success is printed on the terminal. This is a crucial test since if the database is not connected successfully then the whole app is kind of pointless. This is implemented in the TestConnection.java file.

Next, we also tested our project's ruggedness by adding various users and going through with their functionalities. So we simulated how an actual erp might work, where the instructor would provide grades then the student would view it. The student would also want to see the timetable. The admin's maintenance mode must be working correctly and so on. We also identified some bugs due to this and were able to fix them.

Diagrams

Use-Case Lists

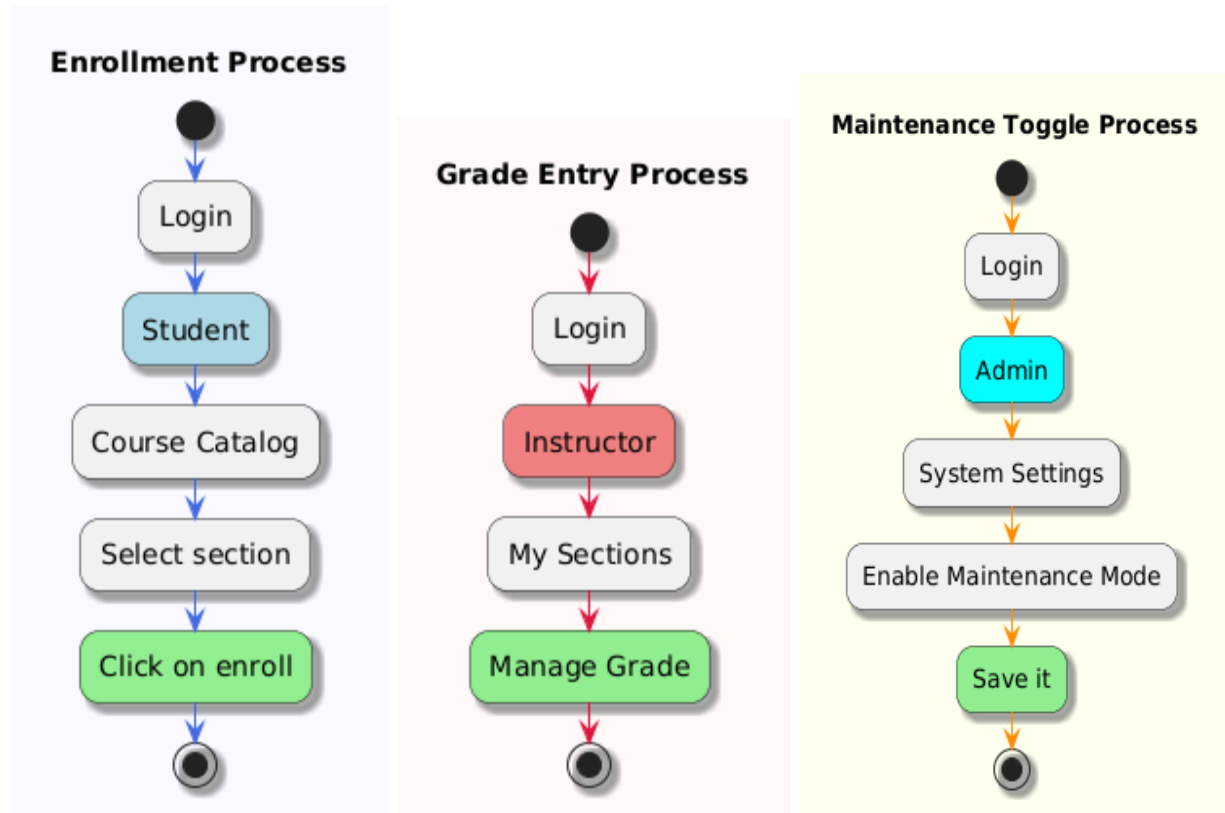
I am assuming that use case lists mean whatever use a user can do. We have 3 roles, so

Student - Login, View Courses, Enroll in Section, Drop a Section, See Enrollments, View Grades, Save grades in CSV or PDF, View TimeTable

Instructor - Login, View Sections, Manage grades, View class statistics, Export grades to CSV/Excel.

Admin - Login, View Users, Add users, View Courses, Delete/Add courses, View sections, Assign instructors, Add sections, Toggle maintenance, Backup/Restore db.

Flow Diagrams



Enroll

Login -> Student -> Course Catalog -> Select section -> Click on enroll

Grade Entry

Login -> Instructor -> My Sections -> Manage Grade

Maintenance Toggle

Login -> Admin -> System Settings -> Enable Maintenance Mode -> Save it

